Sai Vijay Bhaskar Mocherla

Graduate Student (2023 - Present)
Surface Dynamics Group
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EDUCATION

M.S. in Chemistry **August 2020**
University of Rochester, USA
CGPA: 3.72/4.00

M.Sc. in Chemistry **April 2019**
Sri Sathya Sai Institute of Higher Learning (SSSIHL), Prasanthi Nilayam, India
CGPA: 8.2/10.0
Thesis: Effect of Torsional Disorder on Exciton Migration in Conjugated Polymers

B.Sc. (Honors) in Chemistry **April 2016**
Sri Sathya Sai Institute of Higher Learning (SSSIHL), Bangalore, India
CGPA: 7.3/10.0 (Major's CGPA: 7.65/10.0)

WORK EXPERIENCE

Visiting Student, Max Planck Institute of Multidisciplinary Sciences, Goettingen **01/2025 - 03/2025**
Advisor: Prof. Alexander Kandratsenka
- Performed plane-wave DFT calculations for training machine-learning based interatomic potentials (MLIP) to simulate the scattering dynamics of C-atoms on Au(111) surface.
- Worked on developing electronic friction based models to simulate scattering gas atoms from metal surfaces.

Graduate Student, Tata Institute of Fundamental Research, Hyderabad **02/2024 - Present**
Advisor: Prof. Pranav R. Shirhatti
Thesis topic : Dynamics of gas - surface scattering in noble gas metal interfaces.
- Studying inelastic scattering of noble gas atoms (Kr, Ar, Xe) from Au(111) surface.
- Development of fast high voltage switches for charge-particle extraction experiments
- Lab rotation project: Calculated the vibrational excitation probability for preparation of state specific gas-phase reagents using single mode continuous wave (cw) lasers. Additionally, a parallelised C++ program (rap-mb) to perform Monte Carlos simulations of rapid adiabatic passage in molecular beams was also developed.

Junior Research Fellow, Tata Institute of Fundamental Research, Hyderabad **09/2021 - 05/2023**
Advisor: Prof. Raghunathan Ramakrishnan
- Studied the effects of electron correlations on ultrafast electron dynamics in atoms and molecules using time-dependent ab initio methods.
- Developed a method to prepare hybrid Gaussian basis sets optimized for calculating higher harmonic (HHG) spectra using the time-dependent configuration interaction (TDCI) approach.

Freelance Programmer **03/2021 - 08/2021**
- During the COVID-19 second wave in India, I took on freelance programming work to develop html based portfolio webpages and javascript based apps while transitioning between research positions.

updated on : 09/05/2026

Project Student, University of Rochester, New York, USA **05/2020 - 02/2021**

Advisor: Prof. Andrew Jordan, Department of Physics and Astronomy

- Studied low-temperature physics of vortex-matter phases in type-II superconductors to develop quantum refrigeration schemes that use fluxons as heat carriers.

Visiting Student, Tata Institute of Fundamental Research, Mumbai **04/2018 - 06/2018**

Advisor: Prof. Venu Gopal Achanta, Department of Condensed Matter Physics and Material Science

- Worked on design of optical nanostructures with dispersion-less plasmon modes, and their fabrication on gold thin films using e-beam lithography and other cleanroom techniques.
- Studied the emergence of broadband optical transmission in 'plasmonic quasi-crystals' using angle-resolved optical transmission measurements. Assisted in setting up the multi-color pump-probe spectroscopy apparatus to measure the lifetimes of charge carriers(hot electrons) in fabricated optical nanostructures.

TEACHING EXPERIENCE

Teaching Assistant, TIFR Hyderabad **01/2026 - 05/2026**

- Assisted with creation and evaluation of assignments for the Quantum Optics (PHY-462.7) course for graduate students in the winter semester 2026.

Teaching Assistant, Department of Chemistry, University of Rochester **08/2019 - 04/2020**

- Organized demonstrations of experiments and supervised lab sessions for freshman General Chemistry (CHEM-131L) during fall '19 semester.
- Conducted problem solving sessions and discussion workshops as a graduate TA (workshop leader) for Physical Chemistry: Thermodynamics, Statistical Mechanics, and Kinetics (CHEM-252) during spring '20 semester.

School Teacher (during Sabbatical year) **08/2016 - 04/2017**

- Taught school children and organized workshops for teachers in rural schools during my sabbatical year.

TECHNICAL SKILLS

Programming and Software: Python, Fortran, C++ and Mathematica

- Proficient in numpy, scipy, matplotlib, pandas, cython, scikit-learn and pybind11.
- Linux shell scripting, BLAS, LAPACK, CMake
- Computational packages: VASP, Gaussian, NWChem, Psi4, LAMMPS
- CAD and Hardware design: LTSpice, KiCAD and SolidWorks

Experiments:

- Skilled in wet chemistry and clean room techniques, including thin-film deposition, sputtering, etching.
- Trained in design and fabrication of optical nanostructures using e-beam lithography.
- Instrumentation and UHV systems

PUBLICATIONS/PRE-PRINTS

- Mocherla, Sai Vijay, and Raghunathan Ramakrishnan. "Variational augmentation of Gaussian continuum basis sets for calculating atomic higher harmonic generation spectra." arXiv preprint arXiv:2307.00732 (2023).